

Generative Adversarial Networks



Supervised vs Unsupervised Learning

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn a function to map $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, image captioning, etc.

Supervised vs Unsupervised Learning

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn a function to map $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, image captioning, etc.



→ Cat

Classification

[This image is CC0 public domain](#)

Supervised vs Unsupervised Learning

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn a function to map $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, image captioning, etc.



A cat sitting on a suitcase on the floor

Image captioning

Caption generated using [neuraltalk2](#)
Image is [CC0 Public domain](#).

Supervised vs Unsupervised Learning

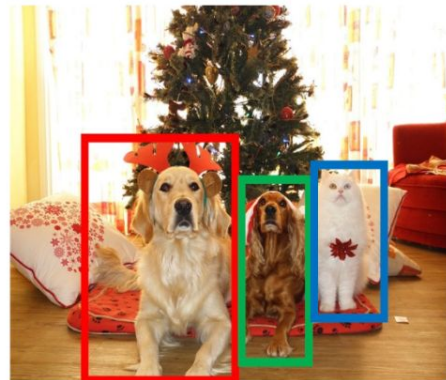
Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn a function to map $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, image captioning, etc.



DOG, DOG, CAT

Object Detection

[This image](#) is [CC0 public domain](#)

Supervised vs Unsupervised Learning

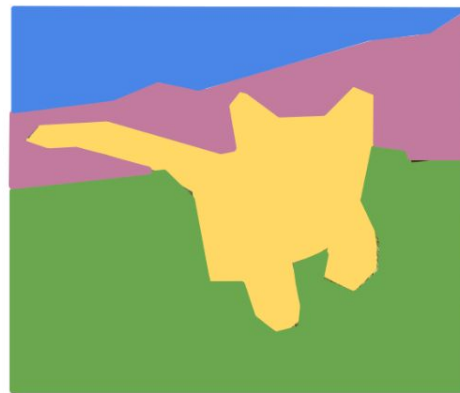
Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn a function to map $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, image captioning, etc.



GRASS, CAT, TREE,
SKY

Semantic Segmentation

Supervised vs Unsupervised Learning

Unsupervised Learning

Data: x

Just data, no labels!

Goal: Learn some underlying hidden structure of the data

Examples: Clustering, dimensionality reduction, density estimation, etc.

Supervised vs Unsupervised Learning

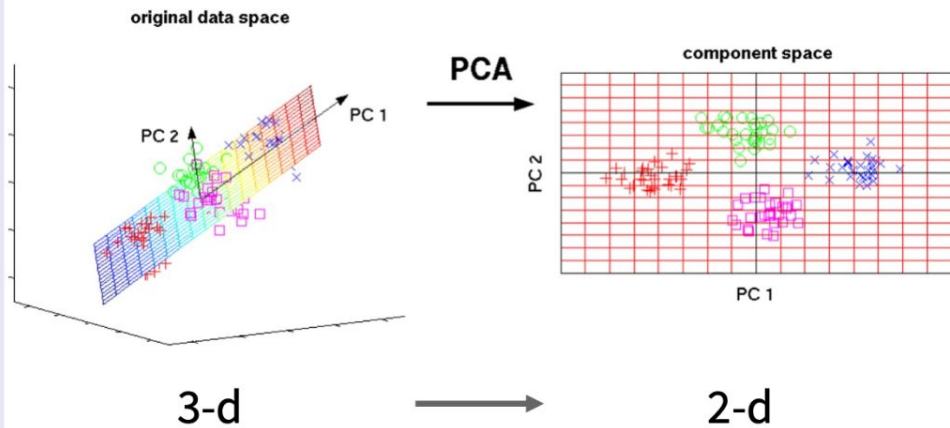
Unsupervised Learning

Data: x

Just data, no labels!

Goal: Learn some underlying hidden structure of the data

Examples: Clustering, dimensionality reduction, density estimation, etc.



Principal Component Analysis
(Dimensionality reduction)

[This image](#) from Matthias Scholz is
[CC0 public domain](#)

Supervised vs Unsupervised Learning

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn a function to map $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, image captioning, etc.

Unsupervised Learning

Data: x

Just data, no labels!

Goal: Learn some underlying hidden structure of the data

Examples: Clustering, dimensionality reduction, density estimation, etc.

Supervised vs Unsupervised Learning

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn a function to map $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, image captioning, etc.

Unsupervised Learning

Data: x

Just data, no labels!

Goal: Learn some underlying hidden structure of the data

Examples: Clustering, dimensionality reduction, density estimation, etc.

Self-Supervised Learning

Data: $(x, \text{pseudo generated } y)$

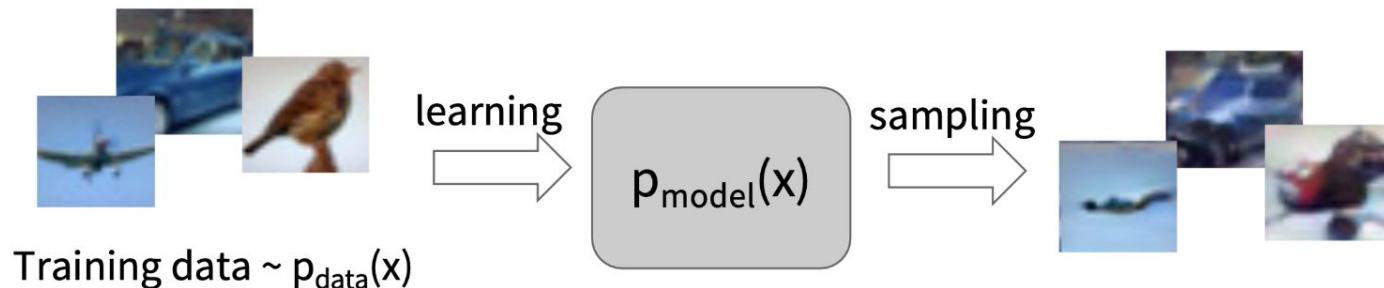
No manual labels!

Goal: Learn to generate good features (reduce the data to useful/generic features)

Example: Classification in downstream applications where we have limited data

Generative Modeling

Given training data, generate new samples from same distribution

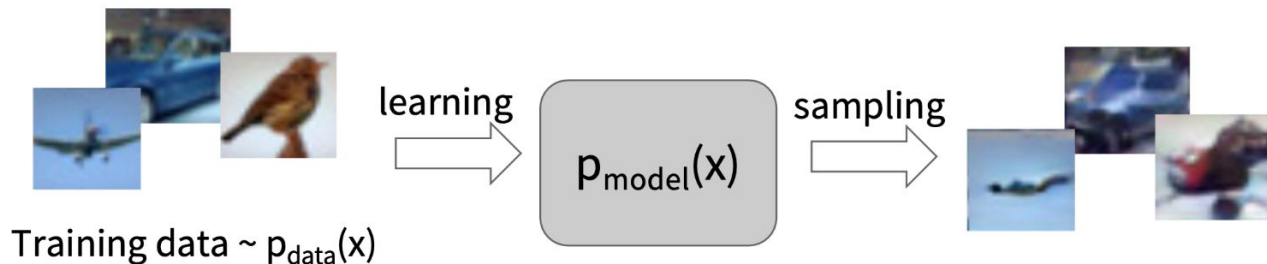


Objectives:

1. Learn $p_{\text{model}}(x)$ that approximates $p_{\text{data}}(x)$
2. Sampling new x from $p_{\text{model}}(x)$

Generative Modeling

Given training data, generate new samples from same distribution



Formulate as density estimation problems:

- Explicit density estimation: explicitly define and solve for $p_{\text{model}}(x)$
- Implicit density estimation: learn model that can sample from $p_{\text{model}}(x)$ without explicitly defining it.

Generative Models

X - tensor in data space

W - learnable parameters of models

p(x|w) - unconditional generation

p(x|c, w), where **c** - conditioning on smth

Generative Adversarial Networks (GANs)

z - latent space

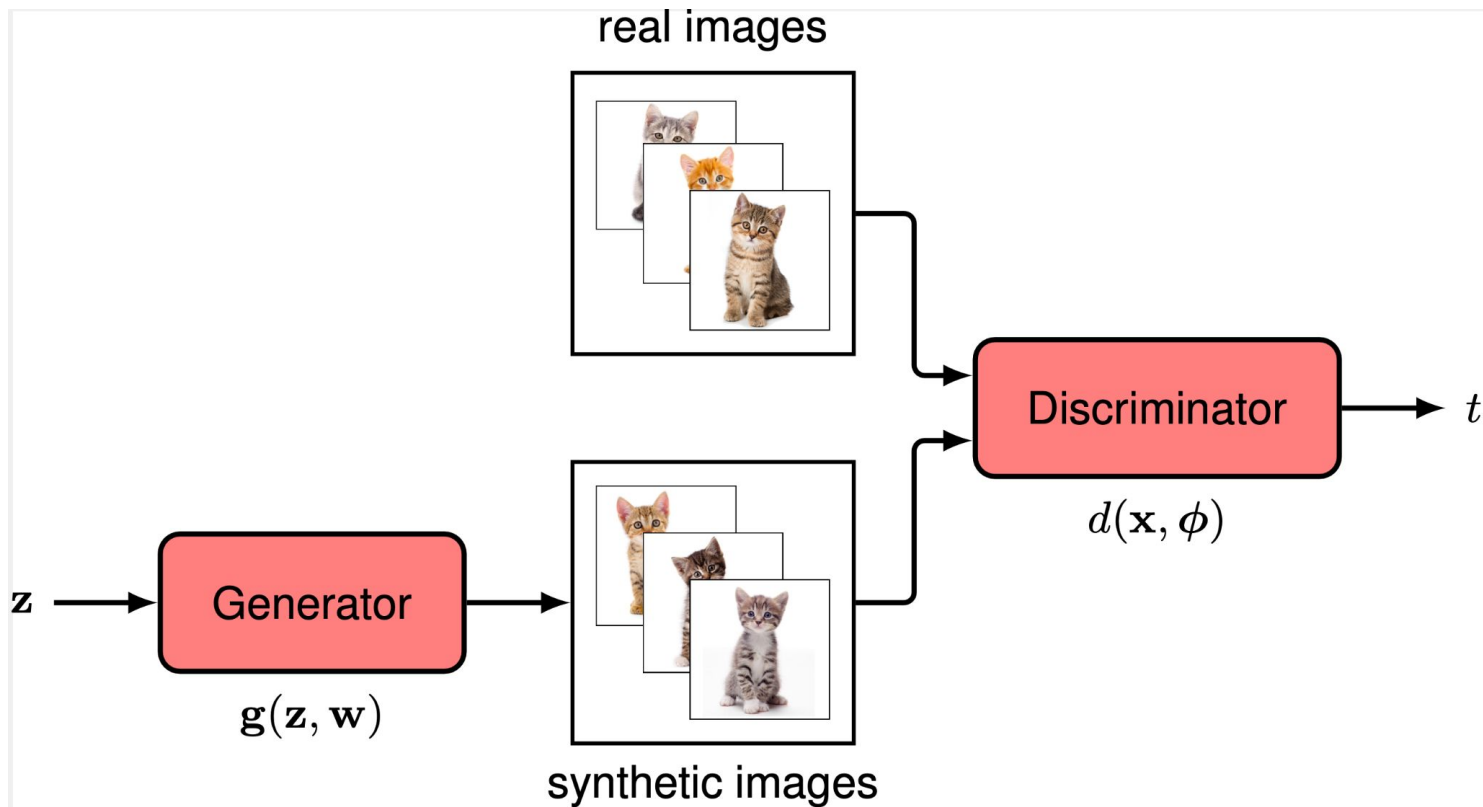
p(z) - Gaussian: **p(z) = N(z|0, I)**

x = g(z, w) - generator

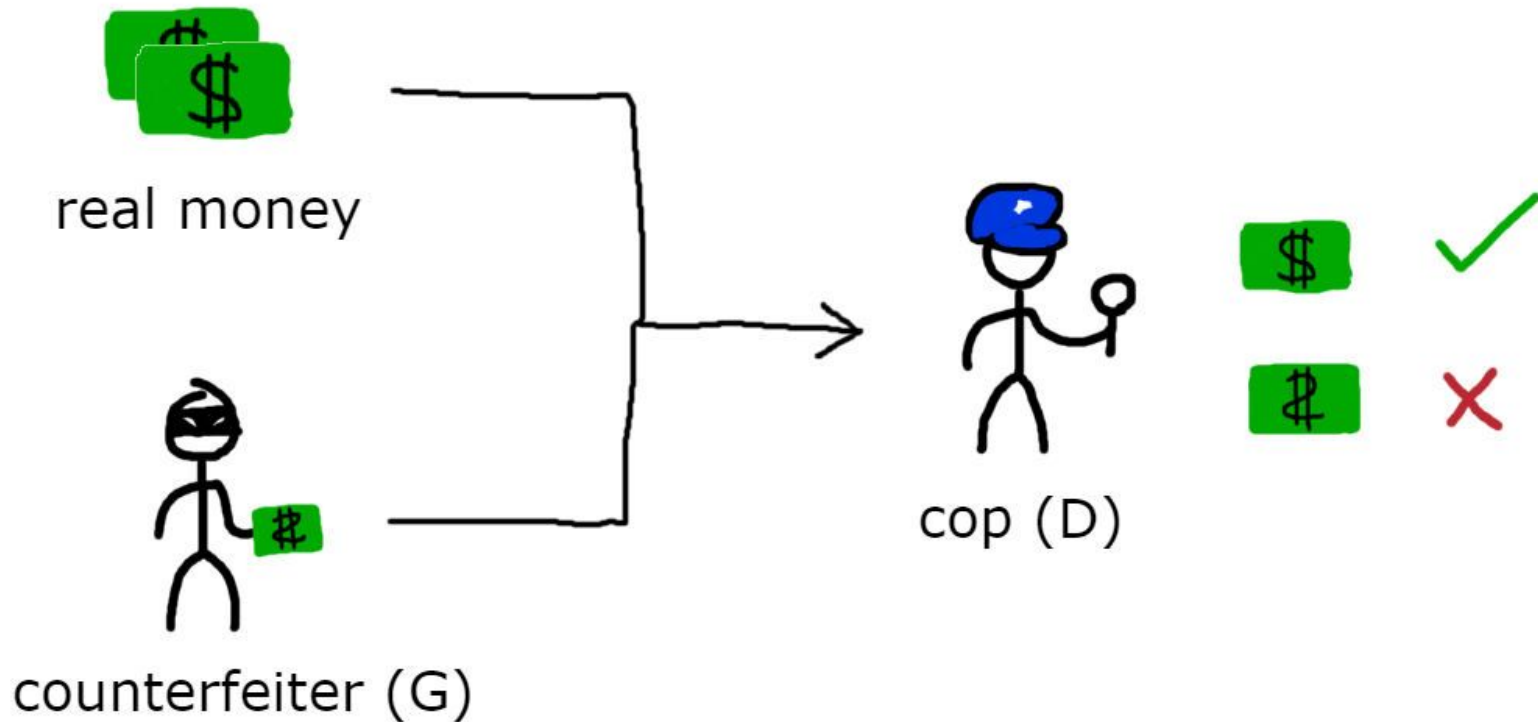
Implicitly defined distribution over **x**

Fit this to data distribution **{x_n}**

Generative Adversarial Networks (GANs)



Generative Adversarial Networks (GANs)



Generative Adversarial Networks (GANs)

$t = 1$, real data

$t = 0$, fake data

$P(t = 1) = d(x, \phi)$ - logistic-sigmoid activation

$$E(\mathbf{w}, \phi) = -\frac{1}{N} \sum_{n=1}^N \{t_n \ln d_n + (1 - t_n) \ln(1 - d_n)\}$$

$$d_n = d(x_n, \phi)$$

Minmax game

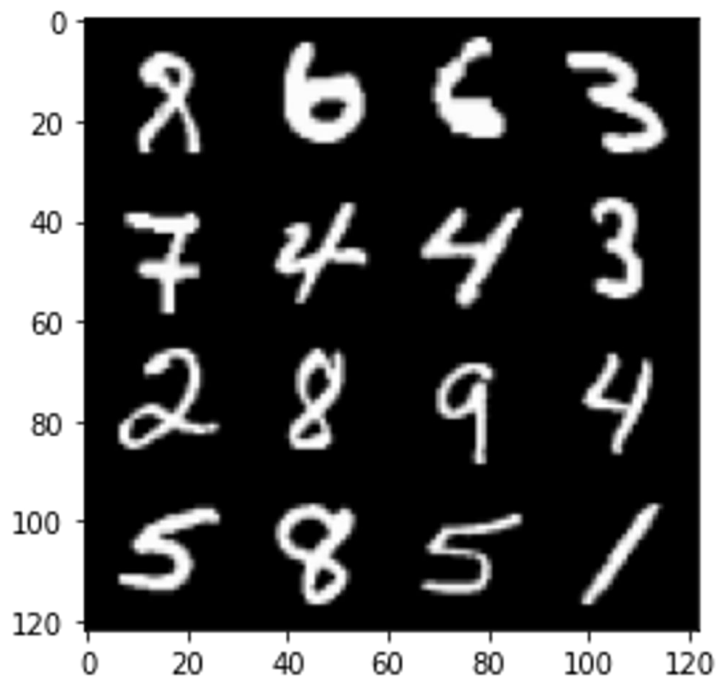
$t_n = 1$ - real

$t_n = 0$ - fake (synthetic)

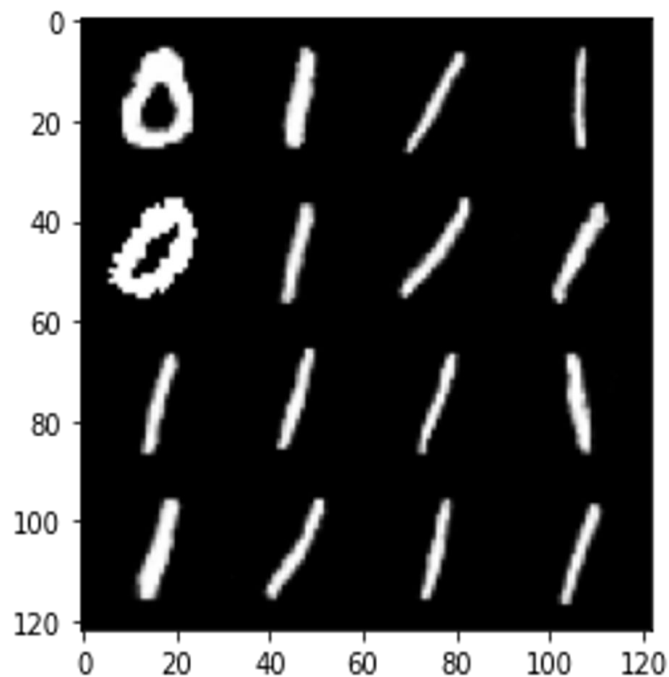
$$E_{\text{GAN}}(\mathbf{w}, \phi) = -\frac{1}{N_{\text{real}}} \sum_{n \in \text{real}} \ln d(\mathbf{x}_n, \phi) \\ - \frac{1}{N_{\text{synth}}} \sum_{n \in \text{synth}} \ln(1 - d(\mathbf{g}(\mathbf{z}_n, \mathbf{w}), \phi))$$

Minimize with respect to ϕ maximized with respect to w

Mode collapse



Train Data Point



GAN Generated Data Point

Simple trick

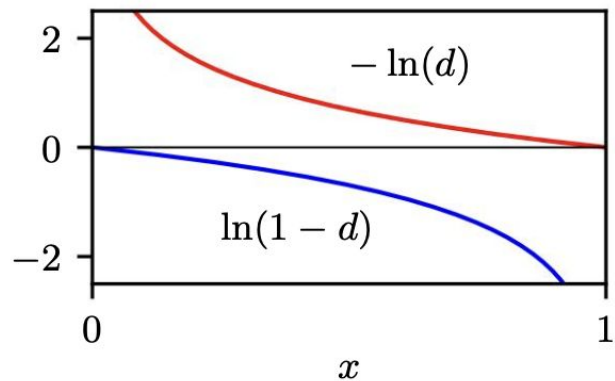
$$-\frac{1}{N_{\text{synth}}} \sum_{n \in \text{synth}} \ln(1 - d(\mathbf{g}(\mathbf{z}_n, \mathbf{w}), \phi))$$

$$\frac{1}{N_{\text{synth}}} \sum_{n \in \text{synth}} \ln d(\mathbf{g}(\mathbf{z}_n, \mathbf{w}), \phi).$$

Simple trick

$$-\frac{1}{N_{\text{synth}}} \sum_{n \in \text{synth}} \ln(1 - d(\mathbf{g}(\mathbf{z}_n, \mathbf{w}), \phi))$$

$$\frac{1}{N_{\text{synth}}} \sum_{n \in \text{synth}} \ln d(\mathbf{g}(\mathbf{z}_n, \mathbf{w}), \phi).$$



Let's create our first GAN!

Metrics?

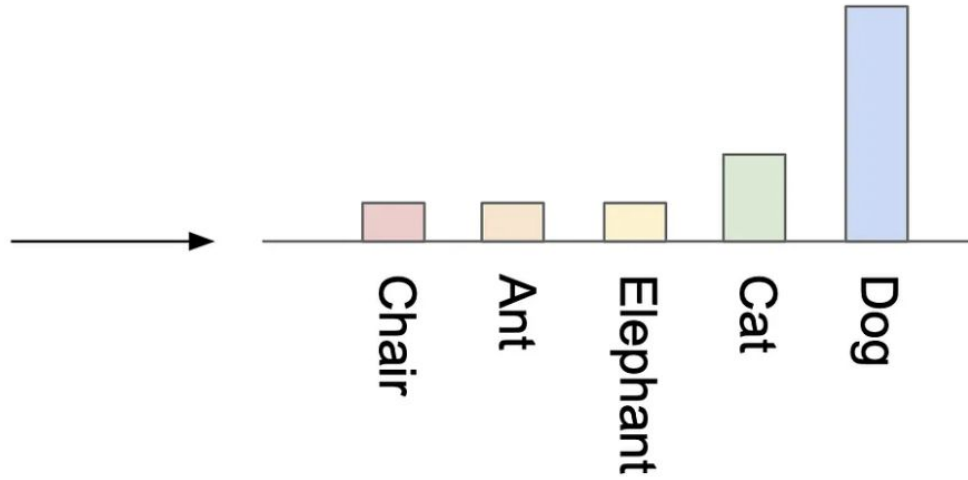


Inception Score

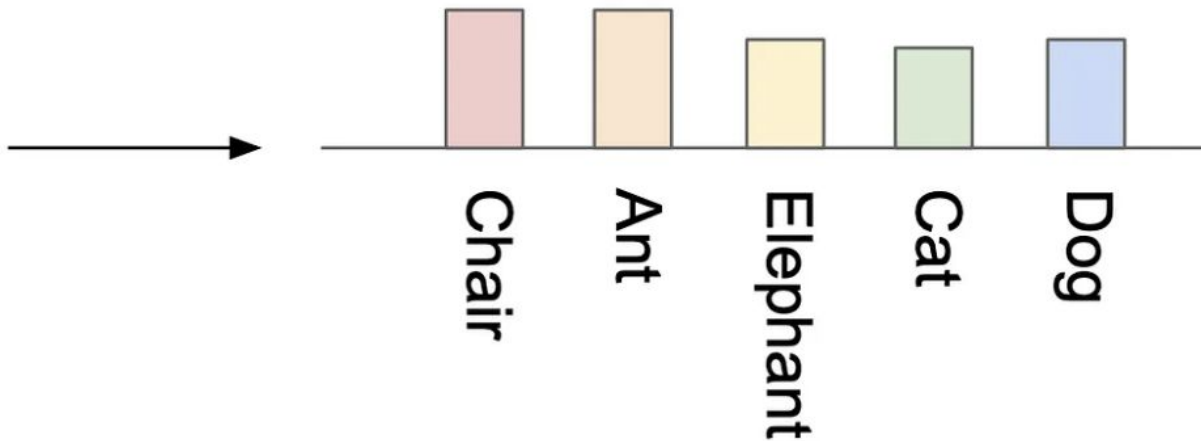
What we want?

- Generated images have variety
- Each image distinctly looks like something

Inception Model

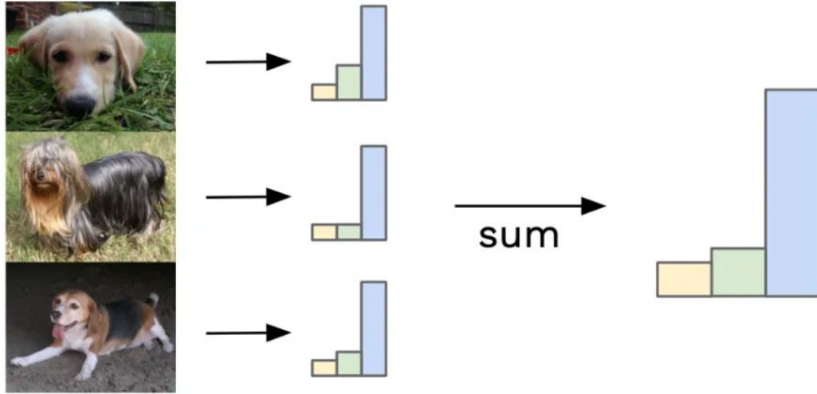


Inception Model

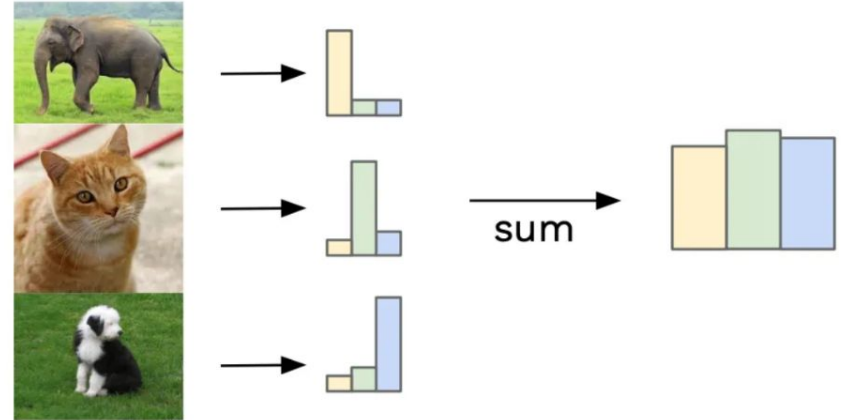


Inception Model

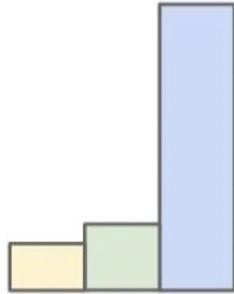
Similar labels sum to give focussed distribution



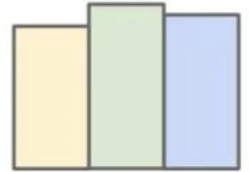
Different labels sum to give uniform distribution



Inception Model



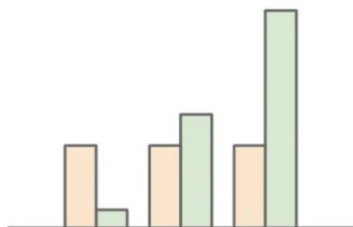
Ideal label distribution



Ideal marginal distribution

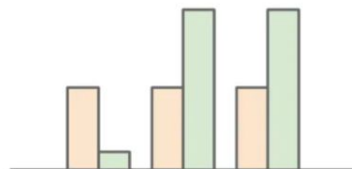
Inception Score

High KL divergence



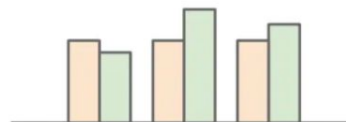
Ideal situation

Medium KL divergence



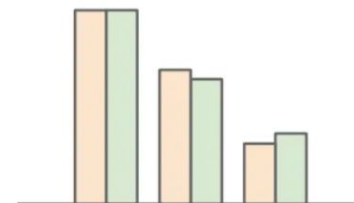
Generated images are
not distinctly one
label

Low KL divergence



Generated images are
not distinctly one
label

Low KL divergence



Generator lacks
diversity

Label distribution

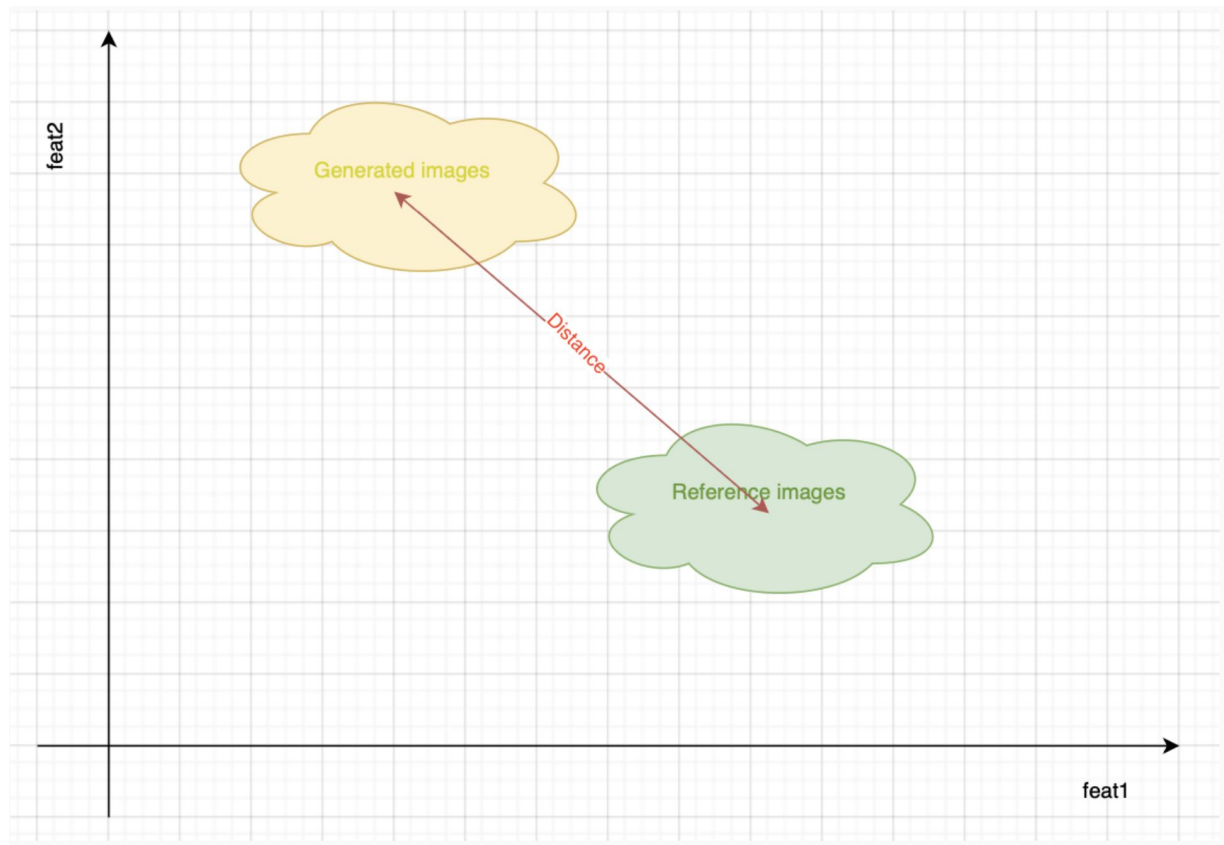
Marginal distribution

Cons?

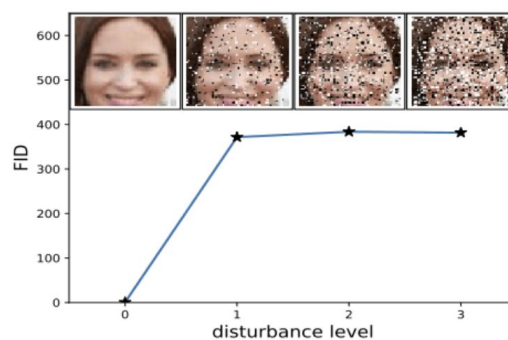
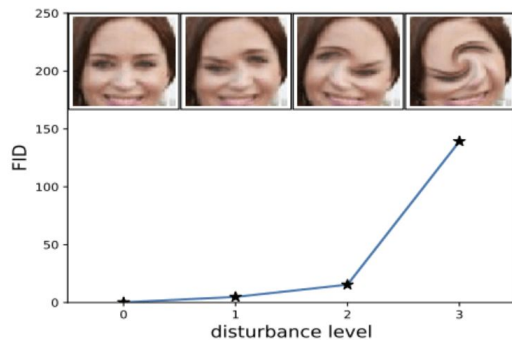
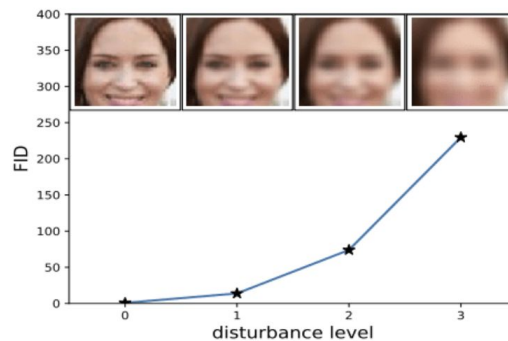
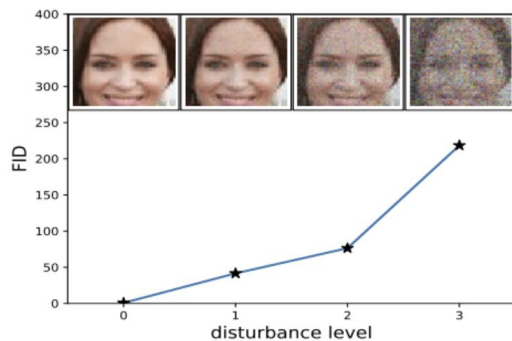
Cons?

- Pre-trained network
- Ignores visual realism
- Mode collapse
- Dataset-specific

Frechet Inception Distance (FID)



Frechet Inception Distance (FID)



Cons of IS?

- Pre-trained network
- Ignores visual realism
- Mode collapse
- Dataset-specific

Cons of IS?

- Pre-trained network
- Ignores visual realism
- Mode collapse
- Dataset-specific

New cons of FID?

New cons of FID?

- Can't measure manipulated or conditioned images

Can we measure manipulations with FID?